

Neal Chandra

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SKILLS SUMMARY

- **Programming Languages/Tools:** Java, C, C++, Python, Verilog, SystemVerilog, x86, RISC-V, Git, Linux
- **Embedded Systems:** Firmware Development, Interrupts, DMA, Low Power Design, UART, SPI, I2C, CAN, PWM, ADC/DAC, BLE
- **AI/ML and Signal Processing:** Tensorflow, Pytorch, Neural Networks, Transformer Architecture, Reinforcement Learning, Time Series Classification, Model Quantization, TinyML, Kalman Filters, Sensor Fusion, Feature Extraction

EDUCATION

- **Arizona State University** Aug 2025 - May 2027
Master of Science; Computer Science GPA: 3.50
- **San Jose State University** Aug 2021 - May 2025
Bachelor of Science; Computer Engineering GPA: 3.60

RESEARCH

- **Multi-Task Unified Vision Transformer for Chest Radiographs**
Used: Python, PyTorch, Numpy, Numba, JIT, Matplotlib, Plotly
 - Created a vision transformer trained across the MIMIC_CXR_JPG, NIH_CHEST_XRAY, ChestX-det datasets
 - Single framework for age regression, orientation, disease localization, classification, segmentation tasks
 - Outperformed state of the art models like DINO and DETR for each task individually

EXPERIENCE

- **Eridu AI** Saratoga, CA
ASIC Engineer Jun 2026 - Present
 - Created websites to display information about ASIC traces like timing violations, capacitance, and falling edge points
 - Stored all Synopsis/Cadence report information in a SQL database. Enhanced multiprocessing capabilities for efficient computation on SLURM nodes
- **Sun Valley Holdings** San Jose, CA
Lead algorithm engineer/Co-founder Jan 2024 - Present
 - Helped start an investing firm by implementing machine learning trading algorithms, Attracted investor attention
 - Implemented Lorentzian Classification, KNN, Convex Optimization, Brownian Motion, and Regression in Python
 - Used the Alpaca Trading API to create data analysis tools and visualizations using Plotly and Matplotlib
 - Constantly updating backend AI logic by implementing various research papers about algorithmic stock trading
- **Hydrolink Systems** San Jose, CA
Embedded Firmware engineer Jan 2024 - Present
 - Joined an effort to make products that track water consumption, recieved investor attention and investment from SJSU
 - Used an ESP32, water flow sensor to measure flow and power the device, sophisticated mosfet switching circuit
 - Used FreeRTOS to manage electronics and send data to backend

CLUBS/PERSONAL PROJECTS

- **Machine Learning** San Jose, CA
Position: Member/Project Lead September 2023 - Present
 - Participated In Research Paper Discussions involving Attention Layers, Natural Language Processing, KAN
- **Formula SAE** San Jose, CA
Position: Member September 2023 - Present
 - Member of the sensor2can subteam. Used STM32 to interface CAN transceiver with dozens of sensors.
 - Utilized Baremetal Programming and Interrupts, Timers, and Watchdog to improve data bandwidth
- **Real Time Exercise Detection**
Used: ESP32C3 microcontroller, Tensorflow/Tensorflow Lite, MPU6050, Edge AI, Resource Constrained ML
 - Designed an edge AI system that performs real time exercise classification
 - Processed raw accelerometer and gyroscopic data using convolutions
 - Implemented quantization using Tensorflow, negligible accuracy loss
 - Integrated hardware, time series analysis, and AI inference into a unified pipeline
- **PWM Signal Generator**
Used: C, Atmega328p, 74HC595 shift register, AVR-GCC
 - Used bit angle modulation to generate PWM signals using shift registers
 - Baremetal programming, uses interrupts to change each signal. Output can handle 900 PWM signals on 0.5 MHz